



ABBOTSLEIGH

2016
HIGHER SCHOOL CERTIFICATE
Assessment 4
Trial Examination

Student's Name: _____

Student Number:

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Teacher's Name: _____

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen.
- **Board-approved** calculators may be used.
- A reference sheet is provided.
- All necessary working should be shown in every question.
- Make sure your HSC candidate Number is on the front cover of each booklet.
- Start a new booklet for Each Question.
- Answer the Multiple Choice questions on the answer sheet provided.
- If you do not attempt a whole question, you must still hand in the Writing Booklet, with the words '**NOT ATTEMPTED**' written clearly on the front cover.

Total marks - 100

- Attempt Sections I and II.

Section I Pages 3 - 6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section.

Section II Pages 7 - 13

90 marks

- Attempt Questions 11 – 16.
- Allow about 2 hr and 45 minutes for this section.
- All questions are of equal value.

Outcomes to be assessed:

Mathematics Extension 2

A student:

HSC :

- E1** Appreciates the creativity, power and usefulness of Mathematics to solve a broad range of problems.
- E2** Chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings
- E3** Uses the relationship between algebraic and geometric representations of complex numbers and of conic sections
- E4** Uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections and polynomials
- E6** Combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions
- E7** Uses the techniques of slicing and cylindrical shells to determine volumes
- E8** Applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae, to problems
- E9** Communicates abstract ideas and relationships using appropriate notation and logical argument

Mathematics Extension 1

A student:

Preliminary course:

- PE1** Appreciates the role of mathematics in the solution of practical problems
- PE2** Uses multi-step deductive reasoning in a variety of contexts
- PE3** Solves problems involving inequalities, polynomials, circle geometry and parametric representations
- PE4** Uses the parametric representation together with differentiation to identify geometric properties of parabolas
- PE5** Determines derivatives that require the application of more than one rule of differentiation
- PE6** Makes comprehensive use of mathematical language, diagrams and notation for communicating in a wide variety of situations

HSC course:

- HE1** Appreciates interrelationships between ideas drawn from different areas of mathematics
- HE2** Uses inductive reasoning in the construction of proofs
- HE3** Uses a variety of strategies to investigate mathematical models of situations involving projectiles
- HE4** Uses the relationship between functions, inverse functions and their derivatives
- HE5** Applies the chain rule to problems including those involving velocity and acceleration as functions of displacement
- HE6** Determines integrals by reduction to a standard form through a given substitution
- HE7** Evaluates mathematical solutions to problems and communicates them in an appropriate form

SECTION I

10 marks

Attempt Questions 1 – 10

Use the multiple-choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

(A) ☐ (B) ☒ (C) ☐ (D) ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

correct 

1. What is the eccentricity of the ellipse $\frac{x^2}{9} + \frac{y^2}{16} = 1$?

(A) $\frac{7}{16}$ (B) $\frac{\sqrt{7}}{4}$ (C) $\frac{9}{16}$ (D) $\frac{7}{9}$

2. $\int_0^{\frac{\pi}{6}} \sin^3 x \, dx$ is equal in value to:

(A) $\int_0^{\frac{\pi}{6}} \sin x - \sin x \cos^2 x \, dx$ (B) $\int_0^{\frac{\pi}{6}} \frac{\pi}{6} - \sin^3 x \, dx$

(C) $\frac{1}{2} \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \sin^3 x \, dx$ (D) $\int_0^{\frac{\pi}{6}} \cos^3 x \, dx$

3. Diagram A shows the complex number z represented in the Argand plane.

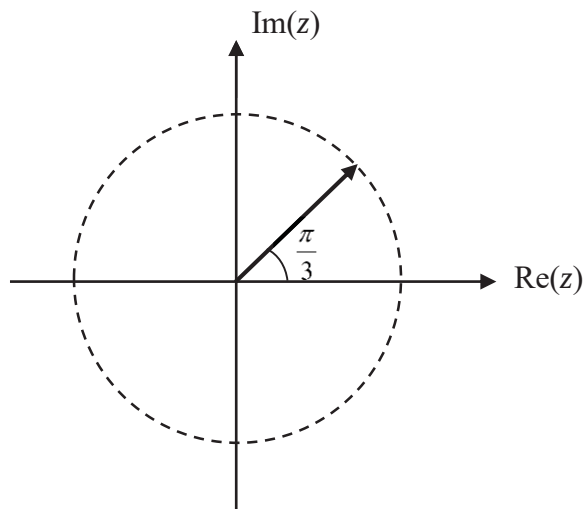


Diagram A

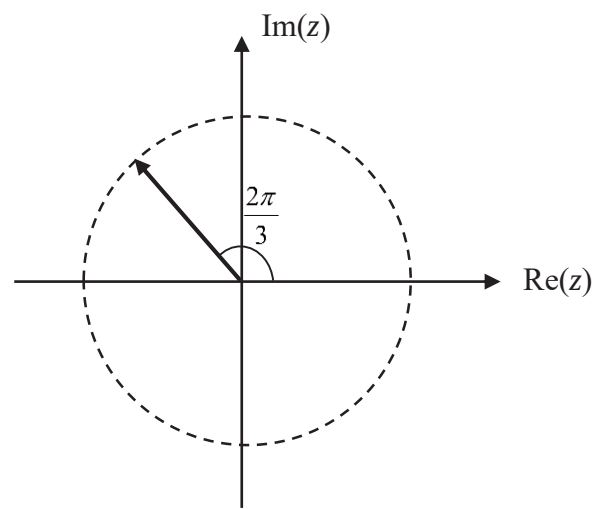
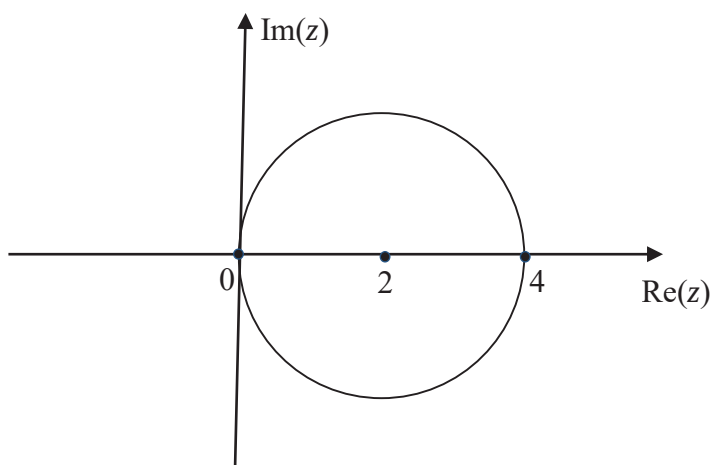


Diagram B

Diagram B shows:

- (A) $\bar{z}$ (B) $2iz$ (C) $-2z$ (D) z^2

4. Which of the following is the equation of the circle below?



- (A) $(z+2)(\bar{z}+2)=4$ (B) $(z-2)(\bar{z}-2)=4$
 (C) $(z+2i)(\bar{z}-2i)=4$ (D) $(z+2)(\bar{z}-2)=4$

5. z and w are two complex numbers. Which of the the following statements is always TRUE?

(A) $|z| - |w| \geq |z + w|$

(B) $|z| + |w| \leq |z - w|$

(C) $|z| + |w| \leq |z + w|$

(D) $|z + w| + |z| \geq |w|$

6. If α , β and γ are the roots of the equation $x^3 - 3x + 4 = 0$.

Then the cubic equation with roots α^2 , β^2 and γ^2 is:

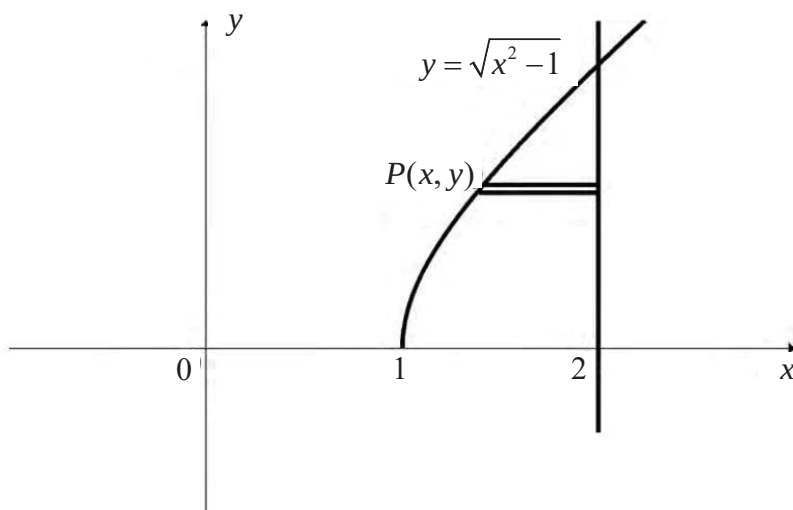
(A) $8x^3 - 9x + 4 = 0$

(B) $x^3 + 9x^2 - 12x + 4 = 0$

(C) $x^3 - 6x^2 + 9x - 16 = 0$

(D) $8x^3 + 4x^2 - 9x + 16 = 0$

7.



The region bounded by the x -axis, the curve $y = \sqrt{x^2 - 1}$ and the line $x = 2$ is rotated about the y -axis.

The slice at $P(x, y)$ on the curve is perpendicular to the axis of rotation.

What is the volume of δV of the annular slice formed?

(A) $\pi(5 - y^2)\delta y$

(B) $\pi(4 - (y^2 + 1))\delta y$

(C) $\pi(4 - x^2)\delta x$

(D) $\pi(2 - x^2)\delta x$

8. What is the acute angle between the asymptotes of the hyperbola $\frac{x^2}{3} - y^2 = 1$?

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$

(D) $\frac{\pi}{2}$

9. Which of the following is the range of the function $f(x) = \sin^{-1} x + \tan^{-1} x$?

(A) $-\pi < y < \pi$

(B) $-\pi \leq y \leq \pi$

(C) $\frac{-3\pi}{4} \leq y \leq \frac{3\pi}{4}$

(D) $\frac{-\pi}{2} \leq y \leq \frac{\pi}{2}$

10. Using the substitution $x = \pi - y$, the definite integral $\int_0^{\pi} x \sin x \, dx$

will simplify to:

(A) 0

(B) $\int_0^{\pi} \sin x \, dx$

(C) $\frac{\pi}{2} \int_0^{\pi} \sin x \, dx$

(D) $\frac{\pi^2}{4}$

End of Section I

SECTION II

Total Marks – 90

Attempt Questions 11 - 16

All questions are of equal value

Answer each question in a **SEPARATE** writing booklet. Extra writing booklets are available.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Given that $z_1 = 5 + 2i$ and $z_2 = 3 - 4i$, find the value of $\operatorname{Re}\left(\frac{z_1}{z_2}\right)$ in $x + iy$ form. 2

(b) (i) Show that the square roots of $-35 + 12i$ are $\pm(1 + 6i)$. 2

(ii) Hence solve $z^2 - (5 + 4i)z + 11 + 7i = 0$. 2

(c) (i) Express $z_1 = -1 + \sqrt{3}i$ in modulus argument form. 2

(ii) Given $z_2 = 3\left(\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right)$, find the value of $z_1 z_2$ in modulus argument form. 2

(iii) Hence express $(z_1 z_2)^3$ in the form $x + iy$, where x and y are real numbers. 2

(d) (i) On an Argand diagram sketch $\arg(z - 2) = \arg z + \frac{\pi}{2}$. 2

(ii) Describe the locus. 1

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

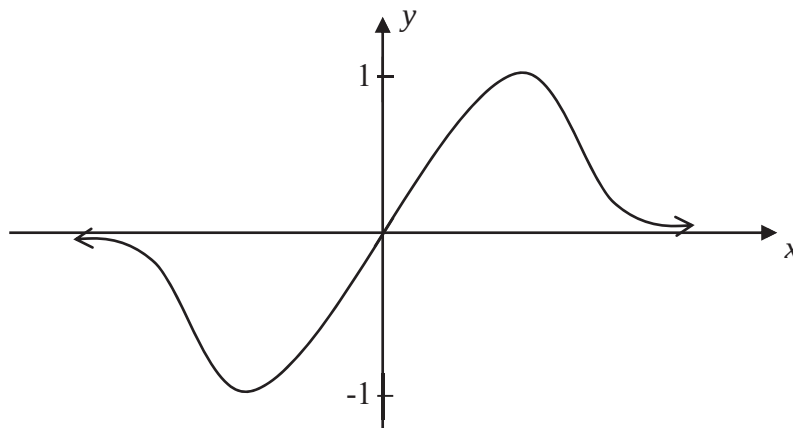
(a) (i) Express $\frac{3x+1}{(x+1)(x^2+1)}$ in the form $\frac{a}{x+1} + \frac{bx+c}{x^2+1}$. 2

(ii) Hence find $\int \frac{3x+1}{(x+1)(x^2+1)} dx$. 2

(b) (i) If $t = \tan \frac{\theta}{2}$, show that $d\theta = \frac{2}{1+t^2} dt$. 1

(ii) By using the substitution $t = \tan \frac{\theta}{2}$, show that $\int_0^{\frac{\pi}{3}} \sec \theta d\theta = \ln(2+\sqrt{3})$. 3

(c) The diagram shows $y = f(x)$ which is an **odd function**. There is a turning point at $(1,1)$.



Draw a separate sketch of each of the following graphs.

Use about one third of a page for each graph. Show all significant features.

(i) $y = \frac{1}{f(x)}$. 2

(ii) $y = f(|x|)$. 1

(iii) $y = e^{f(x)}$. 2

(iv) $y = f(x) \times \sin^{-1} x$ (show the coordinates of the endpoints). 2

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) Given that the roots of the equation $4x^3 - 24x^2 + 45x - 26 = 0$ form an arithmetic sequence, solve the equation. 3
- (b) The polynomial $P(x) = 12x^3 + 44x^2 - 5x - 100$ has a double root. 2
Factorise $P(x)$ over the real number system.
- (c) The hyperbola H has equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and eccentricity e , while the ellipse E has equation $\frac{x^2}{a^2 + b^2} + \frac{y^2}{b^2} = 1$.
- (i) Show that E has eccentricity $\frac{1}{e}$ and hence that E has equation $\frac{x^2}{a^2 e^2} + \frac{y^2}{b^2} = 1$. 2
- (ii) Show that E passes through one focus of H , and H passes through one focus of E . 2
- (iii) Sketch H and E on the same diagram, labelling the foci S, S' of H and T, T' of E , and the directrices of H and E . Give the coordinates of the foci and the equations of the directrices in terms of a and e . 2
- (iv) If H and E intersect at P in the first quadrant. 2
Show that the coordinates of P is $\left(ae\sqrt{\frac{2}{e^2 + 1}}, \frac{a(e^2 - 1)}{\sqrt{1 + e^2}} \right)$.
- (v) Show that the acute angle α between the tangents to the curves at P satisfies $\tan \alpha = \sqrt{2} \left(e + \frac{1}{e} \right)$. 2

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

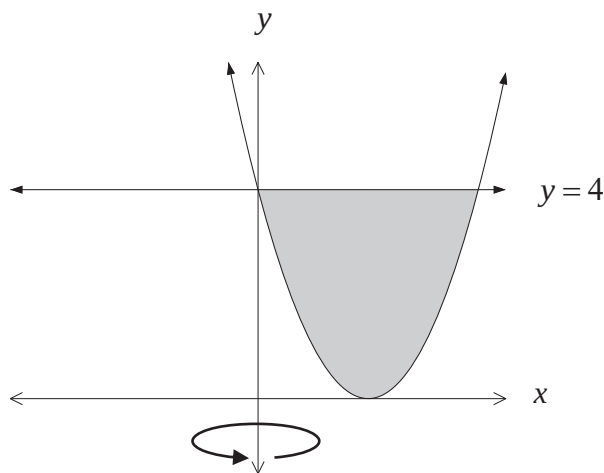
- (a) The equation $x^3 + 2x - 1 = 0$ has roots α, β, γ . Evaluate $\alpha^3 + \beta^3 + \gamma^3$. 2

(b) For $n = 0, 1, 2, \dots$ let
$$I_n = \int_0^{\frac{\pi}{4}} \tan^n \theta \, d\theta.$$

- (i) Show that $I_1 = \frac{1}{2} \ln 2$. 2

- (ii) Use **integration by parts** to show that, for $n \geq 2$, $I_n + I_{n-2} = \frac{1}{n-1}$. 4

- (c) The area enclosed by the curve $y = (x-2)^2$ and the line $y = 4$ is rotated around the y -axis. Use the method of cylindrical shells to find the volume formed. 3



- (d) (i) On the same number plane diagram sketch the curves 2

$$y = |x| - 2 \text{ and } y = 4 + 3x - x^2.$$

- (ii) Hence or otherwise solve the inequality $\frac{|x| - 2}{4 + 3x - x^2} > 0$. 2

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ lie on the rectangular hyperbola $xy = c^2$.

The chord PQ meets the x -axis at C .

O is the centre of the hyperbola and R is the midpoint of PQ .

- (i) Draw a sketch showing all information. 1

- (ii) Find the equation of the chord PQ . 2

- (iii) Find the coordinates of C . 1

- (iv) Find the coordinates of R . 1

- (v) Show that $OR = RC$. 2

- (b) (i) Show that $\sin(A+B) - \sin(A-B) = 2 \cos A \sin B$. 1

- (ii) Use the method of Mathematical Induction to show that for all positive integers n : 4

$$\cos x + \cos 2x + \cos 3x + \dots + \cos nx = \frac{\sin\left(n + \frac{1}{2}\right)x - \sin \frac{1}{2}x}{2 \sin \frac{1}{2}x}$$

- (iii) Hence show that: 3

$$\cos 2x + \cos 4x + \cos 6x + \dots + \cos 16x = 8 \cos 9x \cos 4x \cos 2x \cos x.$$

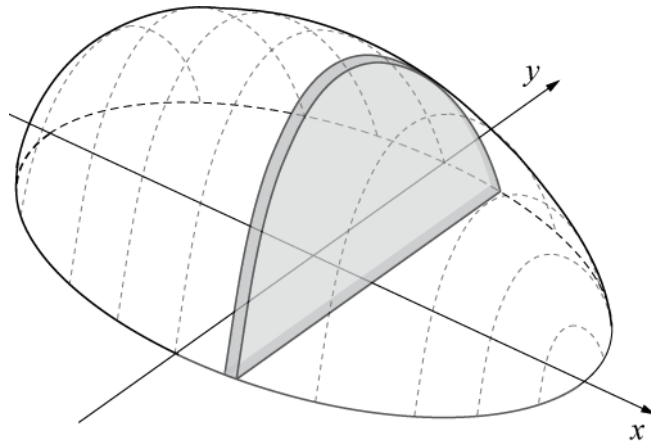
End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Show that the area enclosed between the parabola $x^2 = 4ay$ and its latus rectum is $\frac{8a^2}{3}$ units². 2

A solid figure (as shown below) has the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$ as its base in the xy plane.

Cross-sections perpendicular to the x -axis are parabolas with latus rectums in the xy plane.



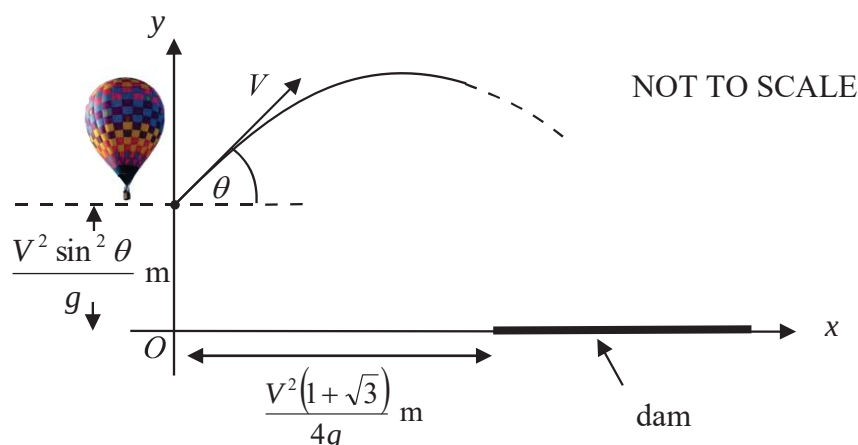
- (ii) Show that the area of the cross-section at $x = h$ is $\frac{16-h^2}{6}$ units². 3

- (iii) Hence, find the volume of this solid. 2

Question 16 continues on the next page.

Question 16 continued.

(b)



A man ascending in a hot air balloon throws a set of car keys to his wife who is on the ground. The keys are projected at a constant velocity of $V \text{ ms}^{-1}$ at an angle of θ to the horizontal, $0^\circ < \theta < 90^\circ$, and from a point $\frac{V^2 \sin^2 \theta}{g}$ m vertically above the ground.

The edge of a dam closest to where the balloon took off, lies $\frac{V^2(1+\sqrt{3})}{4g}$ m

horizontally from the point of projection. The dam is $\frac{V^2}{2g}$ m wide.

The position of the keys at time t seconds after they are projected is given by:

$$\begin{aligned} x &= Vt \cos \theta \\ \text{and} \quad y &= \frac{-gt^2}{2} + Vt \sin \theta + \frac{V^2 \sin^2 \theta}{g} \end{aligned}$$

- (i) Show that the Cartesian equation of the path of the keys is given by: 1

$$y = \frac{-gx^2 \sec^2 \theta}{2V^2} + x \tan \theta + \frac{V^2 \sin^2 \theta}{g}$$

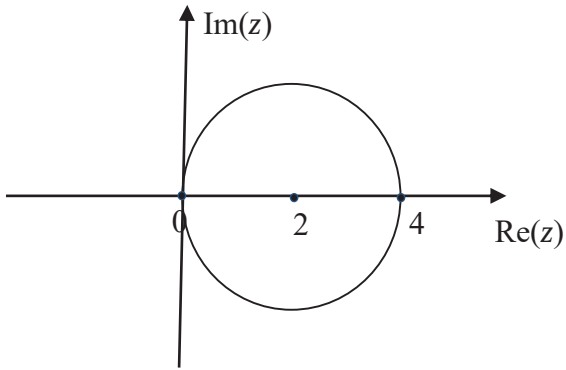
- (ii) Show that the horizontal range of the keys on the ground is given by: 3

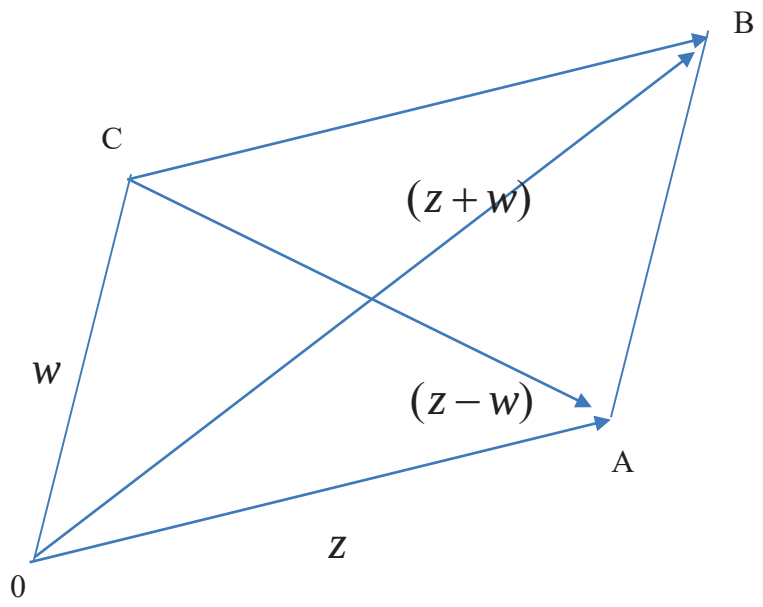
$$x = \frac{V^2(1+\sqrt{3}) \sin 2\theta}{2g}$$

- (iii) Find the values of θ for which the keys will **NOT** land in the dam. 4

END OF PAPER

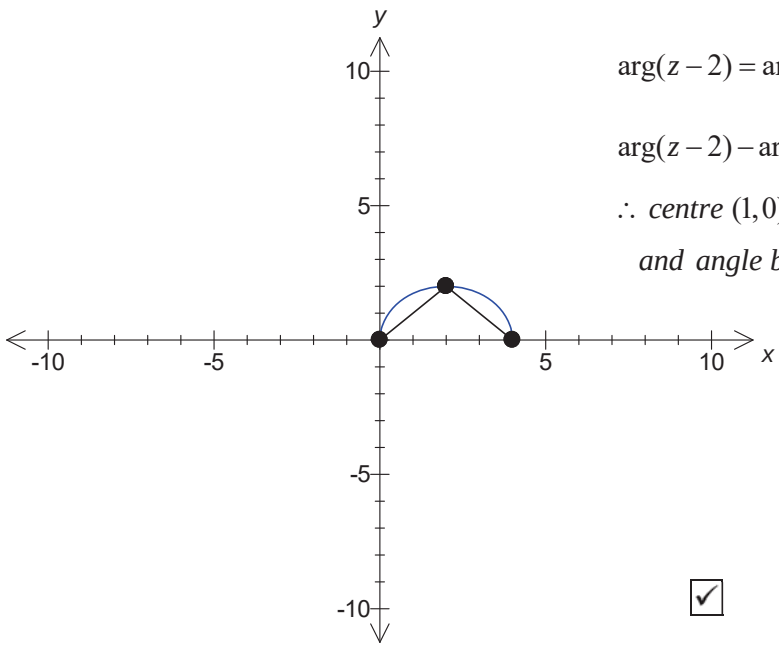
Extension 2 Mathematics Task 4 Trial Examination 2016 Solutions:

Question	Working	Solution
1	$\frac{x^2}{9} + \frac{y^2}{16} = 1$ since $b > a$ $\therefore a^2 = b^2(1 - e^2)$ $9 = 16(1 - e^2)$ $e^2 = \frac{7}{16}$ $e = \frac{\sqrt{7}}{4} \quad \therefore B$	B
2	$\int_0^{\frac{\pi}{6}} \sin^3 x \, dx$ $= \int_0^{\frac{\pi}{6}} \sin x \cdot \sin^2 x \, dx$ $= \int_0^{\frac{\pi}{6}} \sin x (1 - \cos^2 x) \, dx$ $= \int_0^{\frac{\pi}{6}} \sin x - \sin x \cos^2 x \, dx \quad \therefore A$	A
3	z^2 double the argument $\therefore D$	D
4	 Centre $(0, 2)$ and radius of 2 $\therefore (z - 2)(\bar{z} - 2) = 4$ $(x + iy - 2)(x - iy - 2) = 4$ $(x + iy)(x - iy) - 2(x + iy) - 2(x - iy) + 4 = 4$ $x^2 + y^2 - 4x + 4 = 4$ $(x - 2)^2 + y^2 = 2^2 \quad \therefore B$	B

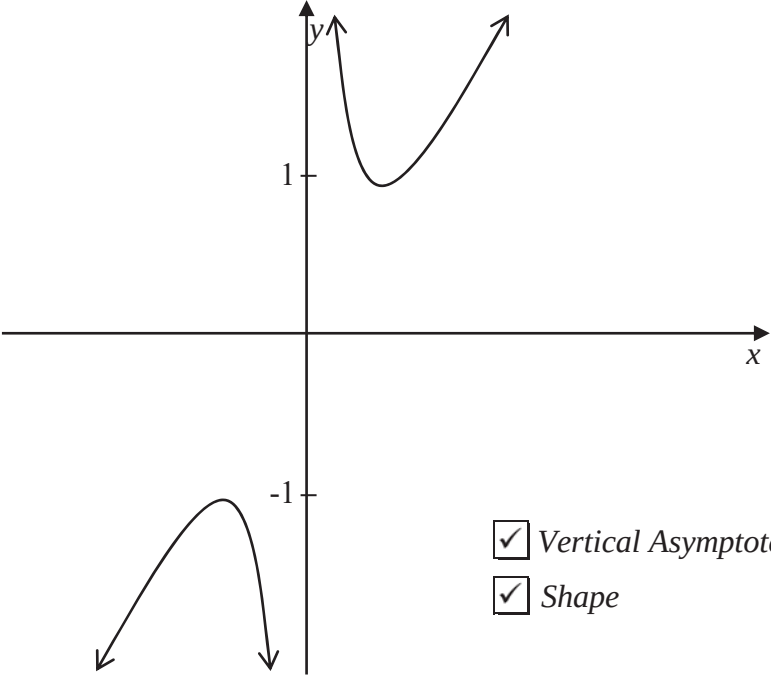
5	 The length of any two sides of a triangle must exceed the length of the third side. Therefore $\Delta OAB \quad z - w \geq z + w \quad \boxed{\times}$ $\Delta OAC \quad z + w \leq z - w \quad \boxed{\times}$ $\Delta AOB \quad z + w \leq z + w \quad \boxed{\times}$ $\Delta AOB \quad z + w + z \geq w \quad \boxed{\checkmark} \quad \therefore D$	D
6	If replace x with $\sqrt{x}$ $(\sqrt{x})^3 - 3(\sqrt{x}) + 4 = 0$ $x\sqrt{x} - 3\sqrt{x} = -4$ $\sqrt{x}(x - 3) = -4$ $x(x - 3)^2 = 16$ $x(x^2 - 6x + 9) = 16$ $x^3 - 6x^2 + 9x - 16 = 0 \quad \therefore C$	C
7	$y = \sqrt{x^2 - 1} \quad \therefore x^2 = y^2 + 1$ $V = \pi(2^2 - x^2)\delta y$ $\therefore V = \pi(4 - (y^2 + 1))\delta y \quad \therefore B$	B

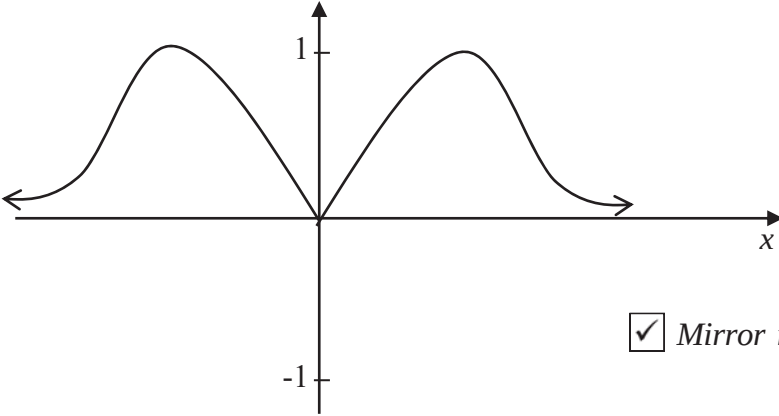
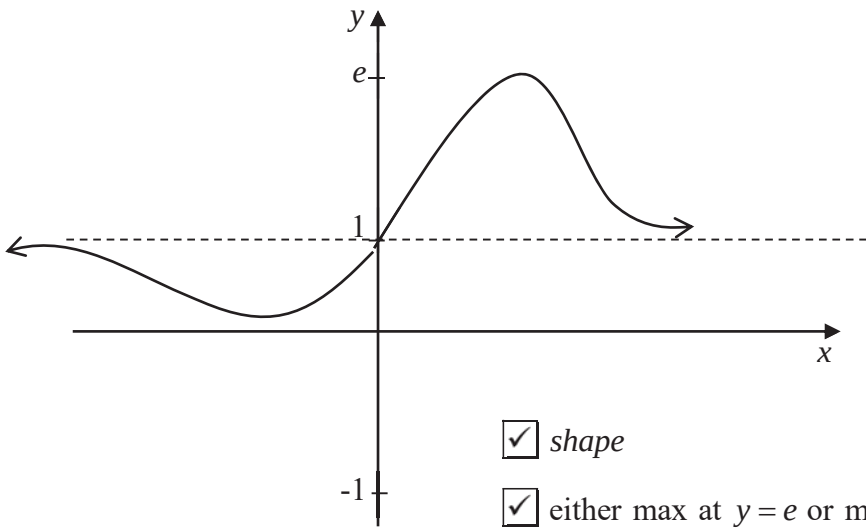
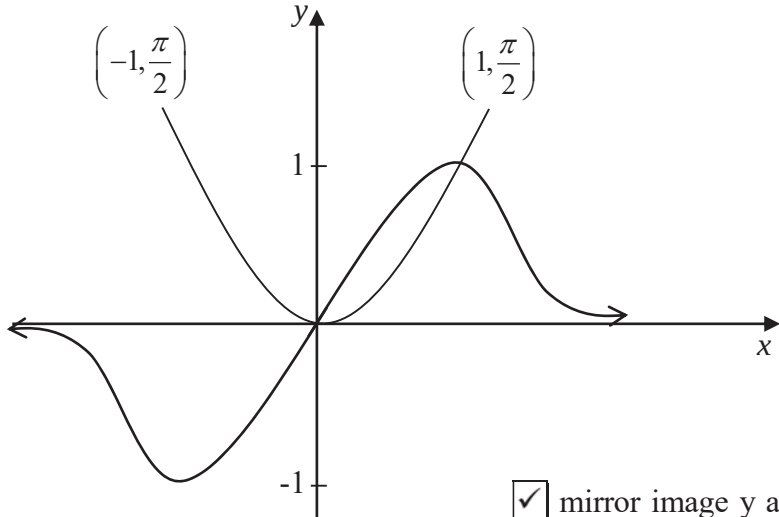
8	Asymptotes $y = \pm \frac{1}{\sqrt{3}} x$ $\therefore \tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right $ $\tan \theta = \left \frac{\left(\frac{1}{\sqrt{3}} \right) - \left(-\frac{1}{\sqrt{3}} \right)}{1 + \left(\frac{1}{\sqrt{3}} \right) \left(-\frac{1}{\sqrt{3}} \right)} \right $ $\tan \theta = \frac{3}{\sqrt{3}}$ $\therefore \theta = 60^\circ = \frac{\pi}{3} \quad \therefore A$	A
9	Domain: $-1 \leq x \leq 1$ $\sin^{-1}(-1) + \tan^{-1}(-1) \leq y \leq \sin^{-1}(1) + \tan^{-1}(1)$ $\frac{-\pi}{2} - \frac{\pi}{4} \leq y \leq \frac{\pi}{2} + \frac{\pi}{4}$ $\frac{-3\pi}{4} \leq y \leq \frac{3\pi}{4} \quad \therefore C$	C
10	$\int_0^\pi x \sin x \, dx = \int_\pi^0 (\pi - y) \sin(\pi - y) \cdot -dy$ $= \int_0^\pi \pi \sin(\pi - y) - y \sin(\pi - y) \, dy$ $= \int_0^\pi \pi \sin y - y \sin y \, dy$ $= \int_0^\pi \pi \sin x - x \sin x \, dx$ $\therefore 2 \int_0^\pi x \sin x \, dx = \pi \int_0^\pi \sin x \, dx$ $\therefore \int_0^\pi x \sin x \, dx = \frac{\pi}{2} \int_0^\pi \sin x \, dx \quad \therefore C$	C

Question	Working	Solution
11(a)	$\left(\frac{z_1}{z_1}\right) = \left(\frac{5+2i}{3-4i}\right)$ $= \left(\frac{5+2i}{3-4i} \times \frac{3+4i}{3+4i}\right)$ $= \left(\frac{15+20i+6i+8i^2}{9-16i^2}\right)$ $= \left(\frac{7+26i}{25}\right) \quad \checkmark$ $\therefore \operatorname{Re}\left(\frac{7+26i}{25}\right) = \frac{7}{25} \quad \checkmark$	2
11(b)(i)	$\sqrt{-35+12i} = a+ib$ $-35+12i = a^2 - b^2 + 2abi$ $\therefore a^2 - b^2 = -35 \quad \text{and} \quad 2ab = 12 \quad (\text{ie: } b = \frac{6}{a})$ $\text{So } a^2 - \left(\frac{6}{a}\right)^2 = -35$ $a^2 - \frac{36}{a^2} = -35 \quad \therefore a^4 + 35a^2 - 36 = 0$ $(a^2 + 36)(a^2 - 1) = 0$ $\therefore a = \pm 6i \quad a = \pm 1 \quad \checkmark$ $\text{So } a = 1 \quad b = 6$ $a = -1 \quad b = -6$ $\text{Therefore } \sqrt{-35+12i} = 1+6i \quad \text{and} \quad -1-6i \quad \checkmark$	2
11(b)(ii)	$z = \frac{5+4i \pm \sqrt{(5+4i)^2 - 4(11+7i)}}{2}$ $z = \frac{5+4i \pm \sqrt{25+40i-16-44-28i}}{2}$ $= \frac{5+4i \pm \sqrt{-35+12i}}{2} \quad \checkmark$ $= \frac{5+4i \pm (1+6i)}{2} \quad \text{from (i)}$ $= 3+5i \quad \text{or} \quad 2-i \quad \checkmark$	2
11(c)(i)	$z_1 = -1 + \sqrt{3}i \Rightarrow z_1 = r(\cos \theta + i \sin \theta)$	

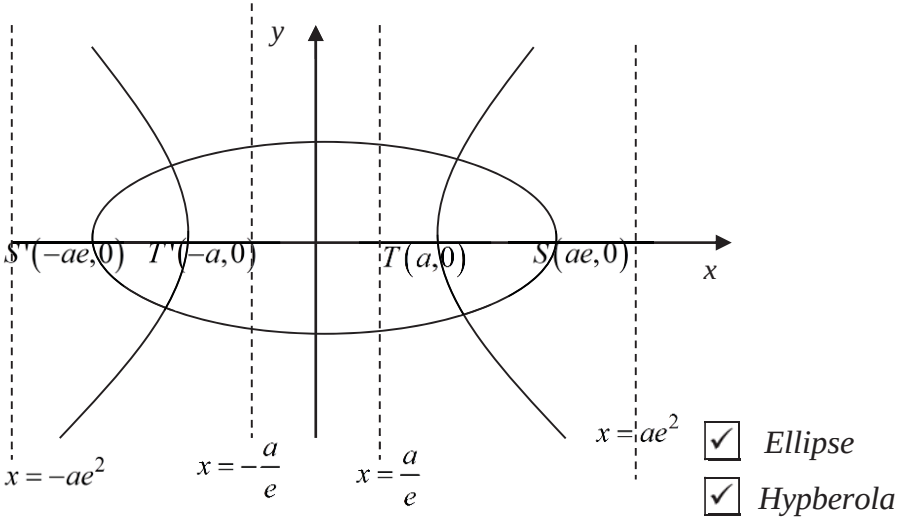
	$\operatorname{mod} z = z \Rightarrow r = \sqrt{(-1)^2 + (\sqrt{3})^2} = 2 \quad \checkmark$ $\arg(z) \Rightarrow \tan \theta = \frac{\sqrt{3}}{1} \therefore \theta = \left(\pi - \frac{\pi}{3}\right) = \frac{2\pi}{3} \quad \checkmark$ $\therefore z_1 = 2\left(\cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right)\right)$	2
11(c)(ii)	$z_1 = 2\left(\cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right)\right) \text{ and } z_2 = 3\left(\cos\left(\frac{\pi}{6}\right) + i \sin\left(\frac{\pi}{6}\right)\right)$ $\therefore z_1 z_2 = \left(2\left(\cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right)\right)\right)\left(3\left(\cos\left(\frac{\pi}{6}\right) + i \sin\left(\frac{\pi}{6}\right)\right)\right)$ $= 6\left(\cos\left(\frac{2\pi}{3} + \frac{\pi}{6}\right) + i \sin\left(\frac{2\pi}{3} + \frac{\pi}{6}\right)\right) \quad \checkmark$ $= 6\left(\cos\left(\frac{5\pi}{6}\right) + i \sin\left(\frac{5\pi}{6}\right)\right) \quad \checkmark$	2
11(c)(iii)	$(z_1 z_2)^3 = \left(6\left(\cos\left(\frac{5\pi}{6}\right) + i \sin\left(\frac{5\pi}{6}\right)\right)\right)^3$ $= 6^3 \left(\cos\left(3 \times \frac{5\pi}{6}\right) + i \sin\left(3 \times \frac{5\pi}{6}\right)\right) \quad \checkmark$ $= 216\left(\cos\left(\frac{5\pi}{2}\right) + i \sin\left(\frac{5\pi}{2}\right)\right)$ $= 216(0 + 1i)$ $= 216i \quad \checkmark$	2
11(d)(i)	 $\arg(z - 2) = \arg z + \frac{\pi}{2}$ $\arg(z - 2) - \arg z = \frac{\pi}{2}$ $\therefore$ centre (1, 0) and (0, 0) and angle between lines is 90° $\checkmark$ $\checkmark$	2
11(d)(ii)	Locus: semi circle with radius 1 and centre (1, 0) $\checkmark$ $\left(\text{ie: } y = \sqrt{1 - (x - 1)^2}\right)$	1

Question	Working	Solution
12(a)(i)	$\frac{3x+1}{(x+1)(x^2+1)} = \frac{a}{x+1} + \frac{bx+c}{x^2+1}$ $a(x^2+1) + (bx+c)(x+1) = 3x+1$ $\therefore \text{Let } x = -1$ $a((-1)^2+1) + (b(-1)+c)((-1)+1) = 3(-1)+1$ $a = -1$ $\therefore ax^2 + bx^2 = 0$ $-1 + b = 0$ $b = 1$ $\therefore x = 0$ $a((0)^2+1) + (b(0)+c)((0)+1) = 3(0)+1$ $a+c=1 \quad c=2$ $\therefore \frac{3x+1}{(x+1)(x^2+1)} = \frac{-1}{x+1} + \frac{x+2}{x^2+1} \quad \checkmark$	2
12(a)(ii)	$\int \frac{3x+1}{(x+1)(x^2+1)} dx = \int \frac{-1}{x+1} + \frac{x+2}{x^2+1} dx$ $= -\ln(x+1) + \int \frac{x}{x^2+1} + \frac{2}{x^2+1} dx \quad \checkmark$ $= -\ln (x+1) + \frac{1}{2} \ln (x^2+1) + 2 \tan^{-1} x + C \quad \checkmark$	2
12(b)(i)	$t = \tan \frac{\theta}{2}$ $\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2}$ $= \frac{1}{2} (1 + \tan^2 \frac{\theta}{2})$ $= \frac{1}{2} (1 + t^2)$ $\therefore \frac{dt}{d\theta} (\tan \frac{\theta}{2}) = \frac{1+t^2}{2} \quad \text{or} \quad d\theta = \frac{2 dt}{1+t^2} \quad \checkmark$ OR $t = \tan \frac{\theta}{2}$ $\tan^{-1} t = \frac{\theta}{2}$ $\theta = 2 \tan^{-1} t$ $\Rightarrow \frac{d\theta}{dt} = \frac{2}{1+t^2}$ $d\theta = \frac{2}{1+t^2} dt$	1

12(b)(ii)	$\left. \begin{aligned} t_1 &= \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}} \\ t_2 &= \tan 0 = 0 \\ \therefore \sec \theta &= \frac{1+t^2}{1-t^2} \end{aligned} \right\} \quad \checkmark$ $= \int_0^{\frac{\pi}{3}} \sec \theta d\theta$ $= \int_0^{\frac{1}{\sqrt{3}}} \frac{1+t^2}{1-t^2} \times \frac{2dt}{1+t^2}$ $= \int_0^{\frac{1}{\sqrt{3}}} \frac{2}{1-t^2} dt$ $= \int_0^{\frac{1}{\sqrt{3}}} \left(\frac{1}{1+t} + \frac{1}{1-t} \right) dt$ $= \left[\ln \left(\frac{1+t}{1-t} \right) \right]_0^{\frac{1}{\sqrt{3}}} \quad \checkmark$ $= \ln \left(\frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} \right)$ $= \ln \left(\frac{\sqrt{3} + 1}{\sqrt{3} - 1} \right) \quad \checkmark$ $= \ln \left(\frac{4 + 2\sqrt{3}}{3 - 1} \right)$ $= \ln(2 + \sqrt{3})$	3
12(c)(i)	 ☒ Vertical Asymptotes and Odd ☒ Shape	2

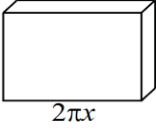
12(c)(ii)	 ☒ Mirror image through y axis	1
12(c)(iii)	 ☒ shape ☒ either max at $y = e$ or min at $y = \frac{1}{e}$	2
12(c)(iv)	 ☒ mirror image y axis and shape ☒ end points	2

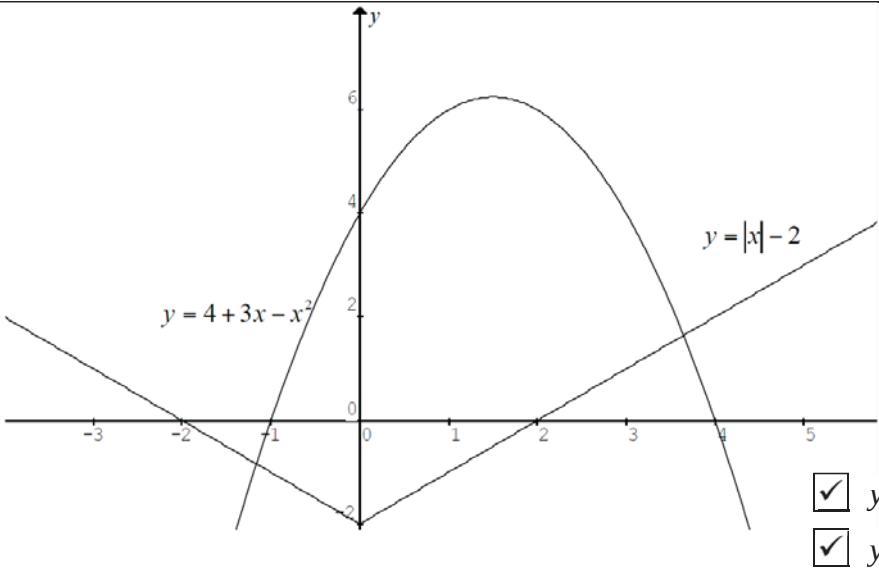
Question	Working	Solution
13(a)	$4x^3 - 24x^2 + 45x - 26 = 0$ $\text{roots } \alpha - d, \alpha, \alpha + d$ $\alpha + \beta + \gamma = -\frac{b}{a} \quad (\alpha - d) + (\alpha) + (\alpha + d) = -\frac{-24}{4} \quad \therefore 3\alpha = 6$ $\alpha = 2$ ☒ $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} \quad \alpha(\alpha - d) + \alpha(\alpha + d) + (\alpha - d)(\alpha + d) = \frac{45}{4}$ $\alpha^2 - \alpha d + \alpha^2 + \alpha d + \alpha^2 - d^2 = \frac{45}{4}$ $3\alpha^2 - d^2 = \frac{45}{4}$ $3(2)^2 - d^2 = \frac{45}{4} \quad \therefore d = \pm \frac{\sqrt{3}}{2}$ ☒ $\therefore \text{Roots } 2 - \frac{\sqrt{3}}{2}, 2, 2 + \frac{\sqrt{3}}{2}$ ☒	3
13(b)	$P(x) = 12x^3 + 44x^2 - 5x - 100$ $P'(x) = 36x^2 + 88x - 5$ $= (18x - 1)(2x + 5)$ ☒ $P(x) = (2x + 5)^2(3x - 4)$ ☒	2
13(c)(i)	For hyperbola $b^2 = a^2(e^2 - 1) \Rightarrow e^2 = \frac{b^2}{a^2} + 1$ or $\frac{b^2 + a^2}{a^2}$ For ellipse let eccentricity be ε. $\left. \begin{aligned} b^2 &= a^2(1 - \varepsilon^2) \\ \text{but } a^2 &= b^2 + a^2 \\ \therefore \frac{b^2}{b^2 + a^2} &= 1 - \varepsilon^2 \\ \varepsilon^2 &= 1 - \frac{b^2}{b^2 + a^2} = \frac{b^2 + a^2 - b^2}{b^2 + a^2} = \frac{a^2}{b^2 + a^2} = \frac{1}{e^2} \\ \therefore \varepsilon &= \frac{1}{e} \end{aligned} \right\}$ ☒ $\left. \begin{aligned} \frac{a^2}{a^2 + b^2} &= \frac{1}{e^2} \\ a^2 + b^2 &= e^2 a^2 \\ \therefore \text{ellipse has equation: } \frac{x^2}{a^2 e^2} + \frac{y^2}{b^2} &= 1 \end{aligned} \right\}$ ☒	2
13(c)(ii)	Focus of hyperbola: $(\pm ae, 0)$	

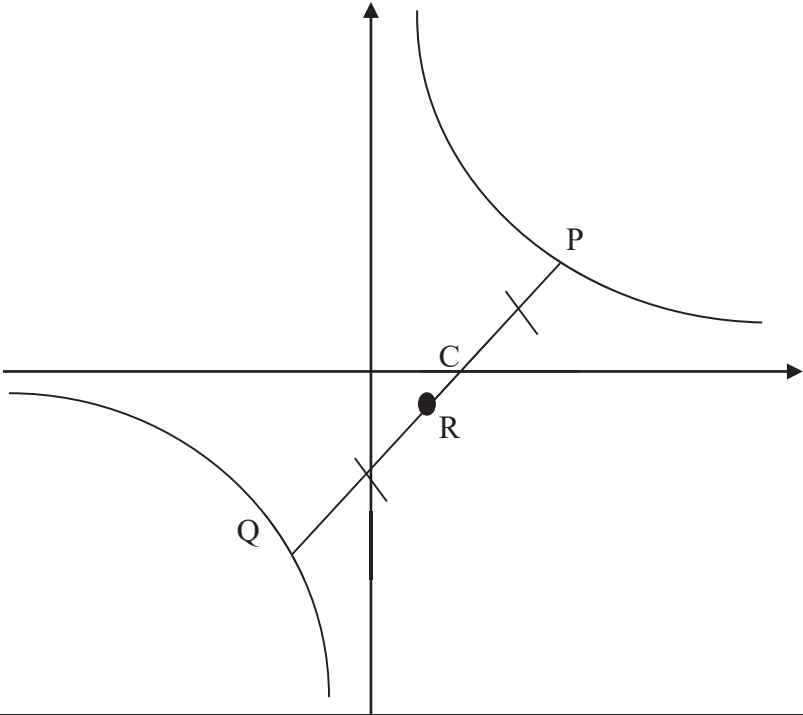
	Focus of ellipse: $(\pm ae, 0)$ but $a = ae$ and $\varepsilon = \frac{1}{e}$ $\therefore \left(\pm ae \left(\frac{1}{e} \right), 0 \right) = (\pm a, 0)$ ☒ Sub $(\pm a, 0)$ into $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $LHS = 1 - 0$ $RHS = 1$ ☒ $\therefore$ the hyperbola passes through $(\pm a, 0)$ Sub $(\pm ae, 0)$ into $\frac{x^2}{a^2 e^2} + \frac{y^2}{b^2} = 1$ $LHS = 1 + 0$ $RHS = 1$ $\therefore$ the ellipse passes through $(\pm ae, 0)$	2
13(c)(iii)	 ☒ Ellipse ☒ Hyperbola	2

13(c)(iv)	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad [1] \quad \frac{x^2}{a^2 e^2} + \frac{y^2}{b^2} = 1 \quad [2]$ $\left. \begin{aligned} \frac{x^2}{a^2} + \frac{x^2}{a^2 e^2} &= 2 \\ x^2(e^2 + 1) &= 2a^2 e^2 \\ x &= \pm \sqrt{\frac{2a^2 e^2}{(e^2 + 1)}} \text{ or } \pm ae \sqrt{\frac{2}{(e^2 + 1)}} \end{aligned} \right\} \quad \checkmark$ $\left. \begin{aligned} \frac{y^2}{b^2} &= 1 - \frac{x^2}{a^2 e^2} \\ &= 1 - \frac{2a^2 e^2}{1 + e^2} \times \frac{1}{a^2 e^2} \\ y^2 &= b^2 \left\{ \frac{e^2 - 1}{1 + e^2} \right\} \quad b^2 = a^2(e^2 - 1) \\ \therefore y^2 &= \frac{a^2(e^2 - 1)^2}{1 + e^2} \\ \therefore y &= \pm \frac{a(e^2 - 1)}{\sqrt{1 + e^2}} \\ \therefore P &\left(ae \sqrt{\frac{2}{e^2 + 1}}, \frac{a(e^2 - 1)}{\sqrt{1 + e^2}} \right) \end{aligned} \right\} \quad \checkmark$	2
13(c)(v)	hyperbola: $\frac{dy}{dx} = \frac{b^2}{a^2} \left(\frac{x}{y} \right)$ ellipse: $\frac{dy}{dx} = \frac{-b^2}{a^2 e^2} \left(\frac{x}{y} \right)$ at P $\frac{dy}{dx} = \sqrt{2}e \quad \frac{dy}{dx} = -\frac{\sqrt{2}}{e} \quad \checkmark$ $\tan \alpha = \left \frac{\sqrt{2}e + \frac{\sqrt{2}}{e}}{1 + \sqrt{2}e \left(\frac{-\sqrt{2}}{e} \right)} \right $ $\tan \alpha = \left \frac{\sqrt{2}e + \frac{\sqrt{2}}{e}}{1 - 2} \right $ $\tan \alpha = \sqrt{2} \left \frac{e + \frac{1}{e}}{-1} \right \quad \checkmark$ $\tan \alpha = \sqrt{2} \left(e + \frac{1}{e} \right)$	2

Question	Working	Solution
14(a)	$\alpha^3 + 2\alpha - 1 = 0 \text{ or } \alpha^3 = -2\alpha + 1 \dots(1)$ $\beta^3 + 2\beta - 1 = 0 \text{ or } \beta^3 = -2\beta + 1 \dots(2)$ $\gamma^3 + 2\gamma - 1 = 0 \text{ or } \gamma^3 = -2\gamma + 1 \dots(3)$ $\therefore (1) + (2) + (3) = \alpha^3 + \beta^3 + \gamma^3$ $= -2\alpha - 2\beta - 2\gamma + 3$ $= 2(\alpha + \beta + \gamma) + 3 \quad \left. \vphantom{\begin{matrix} \alpha^3 + 2\alpha - 1 = 0 \\ \beta^3 + 2\beta - 1 = 0 \\ \gamma^3 + 2\gamma - 1 = 0 \end{matrix}} \right\} \boxed{\checkmark}$ since $(\alpha + \beta + \gamma) = \frac{-b}{a} = 0$ then $2(0) + 3 = 3 \quad \boxed{\checkmark}$	2
14(b)(i)	$I_1 = \int_0^{\frac{\pi}{4}} \tan \theta d\theta = \int_0^{\frac{\pi}{4}} \frac{\sin \theta}{\cos \theta} d\theta = -[\ln(\cos \theta)]_0^{\frac{\pi}{4}} \quad \boxed{\checkmark}$ $= -\left[\ln \frac{1}{\sqrt{2}} - \ln 1 \right]$ $= \ln \sqrt{2} \text{ or } \frac{1}{2} \ln 2 \quad \boxed{\checkmark}$	2
14(b)(ii)	$I_n = \int_0^{\frac{\pi}{4}} \tan^n \theta d\theta = \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \times \tan^2 \theta d\theta$ $= \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \times (\sec^2 \theta - 1) d\theta$ $= \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \times \sec^2 \theta d\theta - \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta d\theta$ $I_n = \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \times \sec^2 \theta d\theta - I_{n-2} \quad (1) \quad \boxed{\checkmark}$	4

	Now $\int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \times \sec^2 \theta d\theta$ ** $u = \tan^{n-2} \theta, u' = (n-2) \tan^{n-3} \theta \times \sec^2 \theta$ $v' = \sec^2 \theta, \quad v = \tan \theta$ $\int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \sec^2 \theta d\theta = \left[\tan^{n-2} \theta \tan \theta \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} (n-2) \tan^{n-2} \theta \sec^2 \theta d\theta \quad \checkmark$ $\therefore \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \times \sec^2 \theta d\theta = 1 - (n-2) \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \times \sec^2 \theta d\theta$ $(1 + (n-2)) \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \times \sec^2 \theta d\theta = 1 \quad \checkmark$ $\int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \times \sec^2 \theta d\theta = \frac{1}{n-1}$ $I_n = \frac{1}{n-1} - I_{n-2} \quad \checkmark$ $\therefore I_n + I_{n-2} = \frac{1}{n-1}$ OR from ** Now $\int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \times \sec^2 \theta d\theta$ $u = \tan^{n-2} \theta, u' = (n-2) \tan^{n-3} \theta \times \sec^2 \theta$ $v' = \sec^2 \theta, \quad v = \tan \theta$ $\int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \sec^2 \theta d\theta = \left[\tan^{n-2} \theta \tan \theta \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} (n-2) \tan^{n-2} \theta \sec^2 \theta d\theta \quad \checkmark$ $\therefore \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \times \sec^2 \theta d\theta = 1 - (n-2) \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \times \sec^2 \theta d\theta$ SO $I_n = 1 - (n-2) \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \times \sec^2 \theta d\theta - I_{n-2}$ BUT $\int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \times \sec^2 \theta d\theta = I_n + I_{n-2}$ from (1) $I_n = 1 - (n-2)(I_n + I_{n-2}) - I_{n-2}$ $I_n + I_{n-2} = 1 - (n-2)(I_n + I_{n-2})$ $(I_n + I_{n-2}) + (n-2)(I_n + I_{n-2}) = 1$ $(I_n + I_{n-2})(1 + n - 2) = 1$ $(I_n + I_{n-2})(n-1) = 1$ $I_n + I_{n-2} = \frac{1}{n-1}$	
14(c)	$V_{Shell} = 2\pi r h \cdot dx$ $= 2\pi x (4 - (x-2)^2) \cdot dx$ $= 2\pi x (4x - x^2) \cdot dx \quad \checkmark$ $V = 2\pi \int_0^4 x (4x - x^2) \cdot dx$ $V = 2\pi \int_0^4 (4x^2 - x^3) \cdot dx$ $V = 2\pi \left[\frac{4x^3}{3} - \frac{x^4}{4} \right]_0^4 \quad \checkmark$ $V = \frac{128\pi}{3} \quad \checkmark$  $4 - y$ $= 4 - (x-2)^2$	3

14(d)(i)	 ☒ $y = x - 2$ ☒ $y = 4 + 3x - x^2$	2
14(c)(ii)	$\frac{ x -2}{4+3x-x^2} > 0$ when $ x -2 > 0$ and $4+3x-x^2$ are both positive or both negative By inspection of the graphs above $2 < x < 4$ ☒ and $-2 < x < -1$ ☒	2

Question	Working	Solution
15(a)(i)		1
15(a)(ii)	$m_{pq} = \frac{\frac{c}{p} - \frac{c}{q}}{cp - cq} = \frac{-1}{pq}$ ☒ Equation PQ: $y - \frac{c}{p} = \frac{-1}{pq} (1 - cp)$ $\Rightarrow x + pqy = c(p + q)$ ☒	2

15(a)(ii)	Let $y = 0$ $x = c(p + q)$ $\therefore C(c(p + q), 0)$ ☒	1
15(a)(iii)	Midpoint : $\left(\frac{cp + cq}{2}, \frac{\frac{c}{p} + \frac{c}{q}}{2} \right) = R \left(\frac{c}{2}(p + q), \frac{c(p + q)}{2pq} \right)$ ☒	1
15(a)(iv)	$OR^2 = \frac{c^2(p + q)^2}{4p^2q^2} + \frac{c^2}{4}(p + q)^2$ $RC^2 = \frac{c^2(p + q)^2}{4p^2q^2} + \left[\frac{c}{2}(p + q) - c(p - q) \right]^2$ $= \frac{c^2(p + q)^2}{4p^2q^2} + \left[-\frac{cp}{2} - \frac{cq}{2} \right]^2$ $= \frac{c^2(p + q)^2}{4p^2q^2} + \left[-\frac{c}{2}(p + q) \right]^2$ $= \frac{c^2(p + q)^2}{4p^2q^2} + \frac{c^2}{4}[(p + q)]^2$ $= OR^2$ $\therefore RC = OR$ ☒	2
15(b)(i)	$\sin(A + B) - \sin(A - B)$ $= \sin A \cos B + \cos A \sin B - \sin A \cos B + \cos A \sin B$ $= 2 \sin A \cos B$ ☒	1
15(b)(ii)	$n = 1$ $LHS = \cos x$ $RHS = \frac{\sin\left(\frac{3}{2}x\right) - \sin\left(\frac{1}{2}x\right)}{2\sin\left(\frac{1}{2}x\right)} = \frac{2\cos x \sin\left(\frac{1}{2}x\right)}{2\sin\left(\frac{1}{2}x\right)} = \cos x$ $\therefore$ true for $n = 1$ ☒	4

Assume true for $n = k$

$$\cos x + \cos 2x + \cos 3x + \dots + \cos kx = \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{1}{2}\right)x}{2\sin\left(\frac{1}{2}\right)x}$$

Prove true for $n = k + 1$

$$RTP: S_{k+1} = \frac{\sin\left(k + \frac{3}{2}\right)x - \sin\left(\frac{1}{2}\right)x}{2\sin\left(\frac{1}{2}\right)x}$$

Now $S_{k+1} = S_k + \cos(k+1)x$

$$= \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{1}{2}\right)x}{2\sin\left(\frac{1}{2}\right)x} + \cos(k+1)x \quad \checkmark$$

$$= \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{1}{2}\right)x + 2\sin\left(\frac{1}{2}\right)x \cos(k+1)x}{2\sin\left(\frac{1}{2}\right)x}$$

$$= \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{1}{2}\right)x + \sin\left(\frac{1}{2}x + (k+1)x\right) - \sin\left((k+1)x - \frac{1}{2}x\right)}{2\sin\left(\frac{1}{2}\right)x} \quad \checkmark$$

$$= \frac{\sin\left(k + \frac{1}{2}\right)x - \sin\left(\frac{1}{2}\right)x + \sin\left(k + \frac{3}{2}\right)x - \sin\left(k + \frac{1}{2}\right)x}{2\sin\left(\frac{1}{2}\right)x}$$

$$= \frac{\sin\left(k + \frac{3}{2}\right)x - \sin\left(\frac{1}{2}\right)x}{2\sin\left(\frac{1}{2}\right)x} \quad \checkmark$$

$\therefore$ true for $n = k + 1$

Now the statement is true for $n = 1$ and we have just proved it true for $n = k + 1$.

$\therefore$ It is true for $n = 1 + 1, i.e n = 2, 3, \dots$

Hence it is true for all positive integers n

15(a)(iii)	$\cos 2x + \cos 4x + \cos 6x + \dots + \cos 16x = \frac{\sin\left(8 + \frac{1}{2}\right)2x - \sin\frac{1}{2}(2x)}{2\sin\frac{1}{2}(2x)}$ $= \frac{\sin 17x - \sin x}{2\sin x}$ $= \frac{\sin(9x + 8x) - \sin(9x - 8x)}{2\sin x} \quad \checkmark$ $= \frac{2\cos 9x \sin 8x}{2\sin x}$ $\sin 8x = 2\sin 4x \cos 4x$ $= 4\sin 2x \cos 2x \cos 4x$ $= 8\sin x \cos x \cos 2x \cos 4x \quad \checkmark$ $\therefore \frac{2\cos 9x \sin 8x}{2\sin x} = \frac{8\cos 9x \sin x \cos x \cos 2x \cos 4x}{\sin x}$ $= 8\cos 9x \cos x \cos 2x \cos 4x \quad \checkmark$	3
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Question	Working	Solution
16(a)	End points of latus rectum are $(2a, a)$ and $(-2a, a)$ $\text{Area} = 4a^2 - \int_{-2a}^{2a} \frac{x^2}{4a} dx \quad \checkmark$ $= 4a^2 - \frac{1}{12a} [x^3]_{-2a}^{2a}$ $= 4a^2 - \frac{8a^2}{6}$ $= \frac{8a^2}{3} \text{unit}^2 \quad \checkmark$	2
16(b)(i)	at $x = h$ $\frac{y^2}{4} = 1 - \frac{h^2}{16}$ $y^2 = 4 - \frac{h^2}{4}$ $y = \pm 2\sqrt{1 - \frac{h^2}{16}} \quad \checkmark$ For parabola $2a = 2\sqrt{1 - \frac{h^2}{16}}$ $a = \sqrt{1 - \frac{h^2}{16}}$ $\therefore \text{Area} = \frac{8}{3} \left(1 - \frac{h^2}{16}\right)$ $= \frac{16 - h^2}{6} \text{units}^2 \quad \checkmark$ OR use Simpson's rule OR $A = 2 \int_0^{2a} a - y \, dx$	3

16(b)(ii)	$\partial V = \frac{16-h^2}{6} \partial h$ $V = \lim_{x \rightarrow \infty} \sum_{h=-4}^4 \frac{16-h^2}{6} \partial h$ $= 2 \int_0^4 \frac{16-h^2}{6} dh \quad \checkmark$ $= \frac{1}{3} \left[16h - \frac{h^3}{3} \right]_0^4$ $= \frac{1}{3} \left[\left(64 - \frac{64}{3} \right) - 0 \right]$ $\text{Volume} = \frac{128}{9} \text{ units}^3 \quad \left(\text{or } 14 \frac{2}{9} \right) \quad \checkmark$	2
16(c)(i)	$\left. \begin{aligned} \text{Given } x &= Vt \cos \theta \quad \text{so } t = \frac{x}{V \cos \theta} \\ \text{in } y &= \frac{-gt^2}{2} + Vt \sin \theta + \frac{V^2 \sin^2 \theta}{g} \\ \text{becomes } y &= \frac{-gx^2}{2V^2 \cos^2 \theta} + \frac{Vx \sin \theta}{V \cos \theta} + \frac{V^2 \sin^2 \theta}{g} \\ \therefore y &= \frac{-gx^2 \sec^2 \theta}{2V^2} + x \tan \theta + \frac{V^2 \sin^2 \theta}{g} \end{aligned} \right\} \quad \checkmark$	1
16(c)(ii)	When, $y = 0$ $y = \frac{-gx^2 \sec^2 \theta}{2V^2} + x \tan \theta + \frac{V^2 \sin^2 \theta}{g}$ $0 = \frac{-gx^2 \sec^2 \theta}{2V^2} + x \tan \theta + \frac{V^2 \sin^2 \theta}{g}$ $x = \frac{-\tan \theta \pm \sqrt{\tan^2 \theta + 4 \times \frac{g \sec^2 \theta}{2V^2} \times \frac{V^2 \sin^2 \theta}{g}}}{\frac{-g \sec^2 \theta}{V^2}} \quad \checkmark$ $= \frac{-\tan \theta \pm \sqrt{\tan^2 \theta + 2 \tan^2 \theta}}{\frac{-g \sec^2 \theta}{V^2}}$ $= \frac{-\tan \theta \pm \sqrt{3 \tan^2 \theta}}{\frac{-g \sec^2 \theta}{V^2}}$ $= \frac{(\tan \theta + \sqrt{3} \tan \theta) V^2}{g \sec^2 \theta} \quad \text{or} \quad \frac{(\tan \theta - \sqrt{3} \tan \theta) V^2}{g \sec^2 \theta}$ $= \frac{V^2 \tan \theta (1 + \sqrt{3})}{g \sec^2 \theta} \quad \text{or} \quad \frac{V^2 \tan \theta (1 - \sqrt{3})}{g \sec^2 \theta}$ $= \frac{V^2 (1 + \sqrt{3}) \sin \theta \cos \theta}{g} \quad \checkmark$ since $1 - \sqrt{3} < 0$ this value of x occurs to the left of the y-axis $\checkmark$ So $x = \frac{V^2 (1 + \sqrt{3}) \sin 2\theta}{2g}$ as required. OR	3

16(c)(ii)	When, $y = 0$ $0 = -\frac{gt^2}{2} + Vt \sin \theta + \frac{V^2 \sin^2 \theta}{g}$ $t = \frac{V \sin \theta \pm \sqrt{V^2 \sin^2 \theta + 4 \times \frac{g}{2} \times \frac{V^2 \sin^2 \theta}{g}}}{2 \times \frac{g}{2}}$ $= \frac{V \sin \theta \pm \sqrt{V^2 \sin^2 \theta + 2V^2 \sin^2 \theta}}{g}$ $= \frac{V \sin \theta \pm \sqrt{3V^2 \sin^2 \theta}}{g}$ $= \frac{V \sin \theta \pm V \sin \theta \sqrt{3}}{g}$ $= \frac{V \sin \theta (1 \pm \sqrt{3})}{g} \quad \text{But } t > 0$ $= \frac{V \sin \theta (1 + \sqrt{3})}{g} \quad \text{sub into } x = Vt \cos \theta$ $x = V \cos \theta \times \frac{V \sin \theta (1 + \sqrt{3})}{g}$ $= \frac{V^2 \sin \theta \cos \theta (1 + \sqrt{3})}{g}$ $= \frac{V^2 \frac{\sin 2\theta}{2} (1 + \sqrt{3})}{g}$ $= \frac{V^2 \sin 2\theta (1 + \sqrt{3})}{g}$	
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16(c)(iii)

From part (ii) the range of the projectile is given by $x = \frac{V^2(1+\sqrt{3})\sin 2\theta}{2g}$

The near edge of the dam is located at $\left(\frac{V^2(1+\sqrt{3})}{4g}, 0 \right)$

At this point, $\frac{V^2(1+\sqrt{3})}{4g} = \frac{V^2(1+\sqrt{3})\sin 2\theta}{2g}$ ☒

$$\therefore \sin 2\theta = \frac{1}{2}$$

$$2\theta = 30^\circ, 150^\circ, \dots$$

$$\theta = 15^\circ, 75^\circ \text{ since } 0^\circ < \theta < 90^\circ$$
 ☒

The far edge of the dam is located at $\left(\frac{V^2(1+\sqrt{3})}{4g} + \frac{V^2}{2g}, 0 \right)$ that is, at $\left(\frac{V^2(3+\sqrt{3})}{4g}, 0 \right)$

At this point,

$$\frac{V^2(3+\sqrt{3})}{4g} = \frac{V^2(1+\sqrt{3})\sin 2\theta}{2g}$$
 ☒

$$3+\sqrt{3} = 2(1+\sqrt{3})\sin 2\theta$$

$$\sin 2\theta = \frac{3+\sqrt{3}}{2+2\sqrt{3}}$$

$$= \frac{3+\sqrt{3}}{2+2\sqrt{3}} \times \frac{2-2\sqrt{3}}{2-2\sqrt{3}}$$

$$= \frac{6-6\sqrt{3}+2\sqrt{3}-6}{4-12}$$

$$= \frac{-4\sqrt{3}}{-8}$$

$$\sin 2\theta = \frac{\sqrt{3}}{2}$$

$$2\theta = 60^\circ, 120^\circ, \dots$$

$$\theta = 30^\circ, 60^\circ \quad 0^\circ < \theta < 90^\circ$$
 ☒